

Ashwin Parameswaran

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SUMMARY

Driven computer science student passionate about AI/ML, data, and software engineering in physical systems. Experienced in applying intelligent models to domains like open-source intelligence, autonomous vehicles, control systems, and embedded platforms. Skilled in research-driven problem solving, technical communication, and development for high-impact engineering.

EDUCATION

Texas A&M University, College Station, TX – *College of Engineering, Class of '26*

Senior Honors Student, Pursuing BS in Computer Science w/ Minor in Mathematics

GPA: 3.66

Financed 100% of college education through merit scholarship awards.

TECHNICAL SKILLS

- **Languages:** C++, Python, Java, PostgreSQL, React.js/Node.js/Express, HTML/CSS, JavaScript, Ruby, MATLAB
- **AI/ML:** PyTorch, TensorFlow, OpenCV; Transformer-based models (LLMs, vision), GPU acceleration (CUDA); privacy-preserving ML (federated/zero-trust); LLM-assisted development (prompt design, agent workflows, AI-TDD)
- **Embedded Systems:** Arduino, STM32/Nucleo, CAN, RTL Logic, PCB Design, Linux, FPGAs & Verilog, Altium
- **Tools:** Jupyter/Colab, Git, Github, Docker, Linux/Unix, VSCode, Ruby on Rails, Excel, HuggingFace, Figma, Jira

EXPERIENCE

Griffiss Institute @ Air Force Research Lab – VICEROY MAVEN Intern:

June 2025 – Aug 2025

- Co-authored & co-developed FROG, (Fragmentation for Obfuscated Geolocation) for DoD open-source intelligence gathering and operations research.
- Designed and authored SCOPE, an DL-powered video analysis pipeline that performs multimodal embedding (BLIP, YAMNet, Sentence-BERT) and semantic video segmentation for public info & intelligence gathering.

Texas A&M Sketch Recognition Lab – Research Project:

June 2024 - Present

- Improved convolutional vision models for medical imaging via segmentation/classification tuning and architecture.
- Brought training time down by 3x and improved model accuracy by an average of 2% across five datasets.
- Collaborated on reproducibility testing and optimization of cutting-edge CV research.

Institute for a Disaster Resilient Texas – Research Project:

September 2024 – Present

- Built GenAI pipelines for classifying disaster imagery using multimodal models (NLP + CV).
- Developed scalable image-to-text analytics for emergency response and impact mapping.
- Co-authored in research poster for computer vision-based elevation estimation for disaster prediction.

Texas A&M Cybersecurity Lab – ICS Engineer

Nov 2023 - Present

- Engineered PLC control logic and anomaly detection models for industrial systems in simulated OT/IT environments
- Developed scenes and control logic for OT environments, researched ML for anomaly detection

ENDEAVR Institute – Autonomous Driving Research Intern:

October 2024 – March 2025

- Developed Python scripts for real-time video capture and preparation, enabling remote autonomous driving
- Part of an agile team converting legacy vehicles into remotely monitored autonomous platforms
- Contributing to a scalable, remotely monitored autonomous rescue and emergency service vehicle platform.

ACTIVITIES

Study Abroad: Swansea University, UK

Jan 2025 – Jun 2025

- Completed CS/Math Coursework while immersed in UK academic paradigm
- Gained international perspective and adaptability through independent travel and academic success abroad

Texas A&M Solar Car Racing Team

Oct 2022 - Present

- Led software/strategy for '26 design cycle, expanding baseline controls to predictive race optimization.
- Overseeing development of solar irradiation, power consumption, vehicle system monitoring, and live telemetry/UI integration with the pits and cockpit to prioritize efficiency rather than only functionality.
- Contributed to CAN bus implementation, PCB design/verification; team placed 7th of 42 at the 2024 American Solar Challenge & Formula Sun Grand Prix.

VICEROY Scholar

Jan 2024 - Present

- Virtual Institutes for Cyber and Electromagnetic Spectrum Research and Employ
- Received stipends for research & coursework in data science, cryptography, and cybersecurity